

Steam Leak Localization Algorithm

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The goal is to determine the position of a steam leak with respect to the robot dog given the following measurements at the robot dog's current and the dog's previous locations:

Current $\mathbf{x}_{D/I_i}^D = [x_{D/I_i}^D, y_{D/I_i}^D]^T$: Translational position of the Dog Frame, D , with respect to the Inertial Frame (stationary observer), I

θ_{I/D_i} : Angular position of the Inertial Frame with respect to the Dog Frame

θ_{L/D_i} : Three speaker measurement of the leak direction with respect to the Dog Frame

Previous $\mathbf{x}_{D/I_{i-1}}^D = [x_{D/I_{i-1}}^D, y_{D/I_{i-1}}^D]^T$: Previous translational position of the Dog Frame with respect to the Inertial Frame

$\theta_{I/D_{i-1}}$: Previous angular position of the Inertial Frame with respect to the Dog Frame

$\theta_{L/D_{i-1}}$: Previous three speaker measurement of the leak direction with respect to the Dog Frame

In the above measurements and throughout this document, the subscripts i and $i - 1$ denote the current and previous measurements taken by the dog, respectively. Figure 1 presents each of the four frames used throughout this derivation along with the repetitive position measurements.

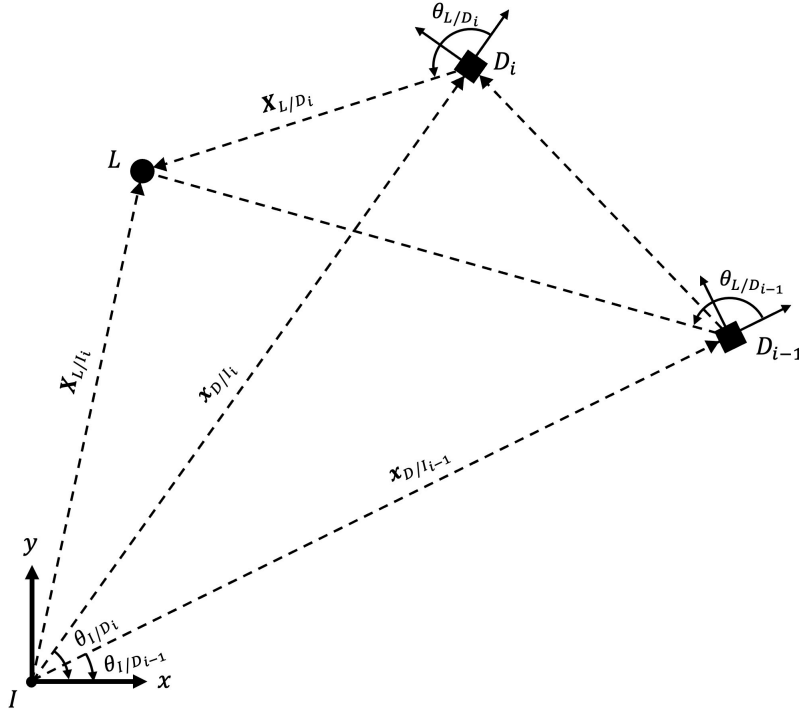


Figure 1: Reference Frames (I : Inertial Frame, L : Steam Leak, D_i : Dog Frame at the i^{th} (current) location, D_{i-1} : Dog Frame at the $i^{th} - 1$ (previous) location)

Subtracting the angular position of the Inertial Frame with respect to the Dog Frame from the three speaker measurement of the leak direction with respect to the Dog Frame, provides the angular position of the leak with respect to the Inertial Frame,

$$\theta_{L/I_i} = \theta_{L/D_i} - \theta_{I/D_i} \quad (1)$$

To find the translational position of the dog with respect to the Inertial Frame resolved in the Inertial Frame, use the transformation:

$$\mathbf{x}_{D/I_i}^I = T_{I/D_i} \mathbf{x}_{D/I_i}^D \quad (2)$$

where $T_{I/D}$ is the rotation matrix,

$$T_{I/D_i} = \begin{bmatrix} \cos(\theta_{I/D_i}) & -\sin(\theta_{I/D_i}) \\ \sin(\theta_{I/D_i}) & \cos(\theta_{I/D_i}) \end{bmatrix} \quad (3)$$

The following equations will estimate the position of the steam leak with respect to a stationary observer, $\mathbf{X}_{L/I}^I = [X_{L/I}^I, Y_{L/I}^I]^T$:

$$X_{L/I}^I = \frac{y_{D/I_i}^I - y_{D/I_{i-1}}^I + |x_{D/I_i}^I| \tan(\theta_{L/I_i}) - |x_{D/I_{i-1}}^I| \tan(\theta_{L/I_{i-1}})}{\tan(\theta_{L/I_{i-1}}) - \tan(\theta_{L/I_i})} \quad (4)$$

$$Y_{L/I}^I = \tan(\theta_{L/I_{i-1}}) X_{L/I}^I + y_{D/I_{i-1}}^I + |x_{D/I_{i-1}}^I| \tan(\theta_{L/I_{i-1}}) \quad (5)$$

Finding the position of the leak with respect to the robot dog,

$$\mathbf{X}_{L/D_i}^I = \mathbf{X}_{L/I}^I - \mathbf{x}_{D/I_i}^I \quad (6)$$

If you want to resolve this in the Dog's Frame, simply use that following transformation,

$$\mathbf{X}_{L/D_i}^D = T_{I/D_i}^T \mathbf{X}_{L/D_i}^I \quad (7)$$

where the superscript T denotes the transpose of the corresponding matrix. After the current measurements, i , are taken make the dog walk in the direction θ_{L/D_i} for a distance of:

$$\text{Walking Distance} = \begin{cases} L_i & \text{if } L_i < 1 \\ 1 & \text{if } L_i \geq 1 \end{cases} \quad \text{where, } L_i = \frac{1}{10} \|\mathbf{X}_{L/D_i}^I\|_2 = \frac{1}{10} \|\mathbf{X}_{L/D_i}^D\|_2 \quad (8)$$

Here, $\|\cdot\|_2$ denotes the Euclidean norm.